

# Dr. Raphael Fischer | Curriculum Vitae

AI Scientist at Lamarr-Institute | TU Dortmund University

Born 10.10.1994. Nationality: German

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To learn about my ongoing activities, visit my [webpage](#) and [LinkedIn](#)

My publications and software are listed on [Google Scholar](#) and [GitHub](#)



My research focuses on the sustainability and trustworthiness of AI, exploring implications for our society, economy, and environment. Guided by interdisciplinarity and real-world applications, my PhD contributed new methods for benchmarking, reporting, labeling, and meta-learning, accompanied by practical solutions from the transfer project with Wilo. I also lead the Young AI Leaders Dortmund, under the UN AI for Good mission.

## Professional Experience

- 07/26 – today**    **Postdoctoral Researcher @ XplaiNLP Group @ Johannes Gutenberg University Mainz**
  - Verifying and extracting disinformation narratives with individualized explanations
  - Coordinating the VeraXtract project and connecting it to my sustainability background
- 06/19 – 06/26**    **AI Scientist @ Lamarr Institute for ML and AI @ TU Dortmund University**
  - Researching sustainability, resource-awareness and trustworthiness of AI
  - 14 peer-reviewed publications, conference and delegation visits, transfer and network activities
- 02/22 – 01/25**    **Data Scientist (part time) @ Wilo SE (Dortmund)**
  - Working in the strategic partnership with Lamarr Institute (research transfer project)
  - Exploring AI use cases and prototyping ML solutions for industrial manufacturing
- 07/16 – 08/18**    **Working Student @ Capgemini (Düsseldorf) & Qiagen (Hilden)**
  - Software development for DNA sequencing, logging and testing with Qiagen as a client

## Education

- 06/19 – 07/25**    **PhD in Computer Science @ TU Dortmund University – Summa cum laude | 1.0 | ECTS A**
  - Topic: Advancing the Sustainability of ML and AI via Labeling and Meta-Learning
  - Committee: JProf. Thomas Liebig, Prof. Geoff Webb, Prof. Christian Janiesch, Prof. Jakob Rehof
- 01/25 – 03/25**    **Research Visit @ Monash University, Melbourne**
  - Exploring resource efficiency in time series classification
- 10/16 – 03/19**    **Master's in Computer Science @ TU Dortmund University – 1.2 | ECTS B**
  - Topic: Filling Gaps in Incomplete Satellite Images with Compressable STRFs
  - Supervisors: Prof. Katharina Morik, Dr. Nico Piatkowski
- 09/18 – 03/19**    **Research Visit @ Monash University, Melbourne**
  - Exploring probabilistic gap filling and cloud removal for satellite image series
- 10/13 – 09/16**    **Bachelor's in Computer Science @ TU Dortmund University – 2.4 | ECTS C**

## Awards & Scholarships

- 2026 YIG Prep Pro Remote Fellow at Karlsruhe Institute of Technology (KIT)**
  - Receiving tailored mentorship, proposal strategy guidance, and training to prepare a junior research group proposal
- 2026 Nominated for the Doctoral Thesis Award from the Gesellschaft für Informatik e.V.**
  - Nominated by the Computer Science Faculty at TU Dortmund University, due to an outstanding PhD
- 2024 Sustainability Award of the Computer Science Faculty @ TU Dortmund University**
  - Awarded for our work on Sustainable LLMs: Energy Efficiency and Transparent Labeling
- 2014 Scholarship from the German Academic Scholarship Foundation (Studienstiftung)**
  - For my Bachelor's and Master's studies, also volunteering as Regional Representative (06/15 – 06/17)

## Teaching Experience

### Supervision

- 01/26 – Master's student assistant Y. Abdelrahim, working on efficiency benchmarking of AI (ongoing)
- 03/26 – MA of T. Horny on Investigating Energy Efficiency Dimensions for Local Deployment in Large Language Models
- 12/25 – BA of B. Demirci on Benchmarking Large Language Models for Sustainable Evaluation
- 10/25 – BA of F. Merece on Interactive AI Labels for Trust: A User-Centric Reporting Framework
- 06/24 – Bachelor's student assistant K. Poitz, working on interdisciplinary evaluations of AI labels (until 05/26)
- 10/24 – BA of F. Theiner on Resource Efficiency of Adversarial Training and Inference of RobustBench Models
- 02/24 – BA of A. Borowski on Energy Efficiency Analysis of Knowledge Graph Embedding Models
- 12/22 – Master's student assistant A. van der Staay, working on benchmarking AI edge devices (until 12/23)
- 11/23 – BA of T. Kubiak on A Comparative Analysis of Parameter-Efficient-Fine-Tuning Methods in LLMs
- 09/22 – MA of F. Dillkötter on Automated Generation of Evaluation Tasks for ML Models

### Lectures and Talks

- 05/26 – Sustainable AI Development (RC-Trust Graduate School)
- 04/26 – Bridging the Communication Gap: Labels for Sustainable AI Development (CEIPI AI Digital Technologies and IP Master)
- 04/26 – Sustainable Development and Energy Efficiency in Deep Learning (ELLIS Mathematics & Efficiency of Deep Learning reading group)
- 10/25 – Sustainable Development in the Age of AI (Computer Science and Sustainability lecture series)
- 10/25 – Fundamentals of Trustworthy AI (Data, Knowledge, and Context in ML and AI lecture series)
- 05/25 – Sustainability and AI (Introduction to Data Science II lecture series)

### Other Activities

- 2025 – Teaching GISSV students (remote) and organizing workshops for Digital Week Dortmund and Girls' Day @ Lamarr
- 2024 – Training course for teachers on AI, Environment, and Sustainability (fobizz / remote)
- 2023 – Organizing workshops for AI4Schools and Digital Week Dortmund @ Lamarr

## Volunteering & Activities

### Young AI Leaders

- Global network that empowers young enthusiasts to drive positive change with AI
- Founded and actively leading the Dortmund Hub, growing to 15 members
- Investigating the use of generative AI for political education in context of the Dortmund municipal elections

### Scientific Community Services

- Editor for Journal of Artificial Intelligence Research (JAIR) Special Track on Challenges in Pretraining 2026
- Reviewing for International Conference on Data Mining (ICDM) 2026
- Reviewing for Conference on Fairness, Accountability, and Transparency (FAccT) 2026

- Reviewing for Explainable AI for Time Series and Data Streams Workshop (TempXAI @ ECML-PKDD) 2025 & 2024
- Reviewing for Conference on AI, Ethics, and Society (AIES) 2025
- Reviewing for Conference on Knowledge Discovery and Data Mining (KDD) 2024
- Reviewing for European Conference on ML and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD) 2022, 2024 & 2026

## **Lamarr Institute**

- Invited talks at ENFIELD Energy & Green AI Workshop (11/25), Sustainable AI Conference (09/25), Greentech.Ruhr meeting (05/25), DOPPLERS (03/25), BAuA KI-Werkstatt (11/24), AISoLA Conference (10/23), and BIFOLD All Hands Meeting (10/23)
- Posters & demos at the Trustworthy AI Summit (09/25), political visits (06/24, 09/23, 08/19), and various strategic events
- Selected delegation member for visits in Canada (10/25), USA (04/24), and France (07/22)
- Research visit & talk at Emma Strubell's Lab @ Carnegie Mellon University, Pittsburgh, USA (02/22)
- Contributed posts to the ML Blog and actively participated in lab visits, scientific forums, pitch talks, and education activities

## **CS Faculty @ TU Dortmund University**

- Sustainability Commission: Conducting and promoting sustainable efforts (06/2025 – Today)
- International Coordination: Assisting students and employees with stays abroad (12/2020 – 12/2025)
- Faculty Council: Representative of the research associates (2022 – 2024)